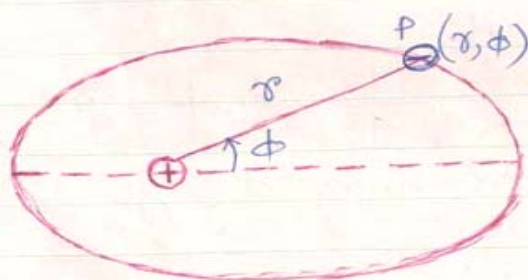




Sommerfeld Theory

The Bohr's theory could not explain the observed fine structure of H α line in the Balmer Series. The two very close wavelengths in H α line suggested that there must be an energy level with slightly different energy values. This splitting of the energy level of slightly different energy values couldn't be explained by Bohr's theory.

Sommerfeld gave a satisfactory explanation by modifying Bohr's circular orbit, to consider the motion of the electron in the elliptical orbit with the nucleus at one of its foci.



Let $P(r, \phi)$ be the position of the electron at any instant t .
The total energy of the electron in the n th orbit:

$$E_n = T_n + V_n$$

$$E_n = \frac{1}{2} m [V_r^2 + V_t^2] + \frac{(+ze) \cdot (-e)}{4\pi\epsilon_0 \cdot r}$$

where $V_r = \dot{r}$ = Radial velocity.

$V_t = r\dot{\phi}$ = Transverse component of velocity



$$1. E_n = \frac{1}{2} m [\dot{r}^2 + r^2 \dot{\phi}^2] - \frac{ze^2}{4\pi\epsilon_0 r} \quad \text{--- (1)}$$

Radial component of momentum = $p_r = m\dot{r}$ --- (2)

The azimuthal or angular momentum

$$p_\phi = I\omega = mr^2\dot{\phi} \quad \text{--- (3)}$$

Since an electron in the elliptical orbit has two degrees of freedom, hence it requires two quantum conditions

$$\oint p_r dr = n_r h \quad \text{--- (4) where } n_r \text{ \& } n_\phi \text{ are integers and are known as Radial quantum number and azimuthal quantum number.}$$

$$\oint p_\phi d\phi = n_\phi h \quad \text{--- (5)}$$

respectively.

Since for a particle moving under a central force (a force which is always directed towards a fixed point), the angular momentum remains constant throughout the motion, p_ϕ is a constant.

from (5) :-

$$p_\phi \oint d\phi = n_\phi h$$

$$\text{or } p_\phi \cdot 2\pi = n_\phi h$$

$$\text{or } p_\phi = \frac{n_\phi h}{2\pi} \quad \text{--- (6)}$$



We now find $\oint p_r dr$:

$$p_r = m\dot{r} = m \frac{dr}{dt} = m \frac{dr}{d\phi} \cdot \frac{d\phi}{dt} = m\dot{\phi} \frac{dr}{d\phi}$$

But from ③ $m\dot{\phi} = \frac{p_\phi}{r^2}$

$$\therefore p_r = \frac{p_\phi}{r^2} \cdot \frac{dr}{d\phi} \quad \text{--- ⑦}$$

$$\text{and } dr = \frac{dr}{d\phi} d\phi \quad \text{--- ⑧}$$

Putting eqns ⑦ and ⑧ in ④

$$\oint \frac{p_\phi}{r^2} \left(\frac{dr}{d\phi} \right) \left(\frac{dr}{d\phi} \right) d\phi = n_r h$$

$$\text{or } p_\phi \oint \left(\frac{1}{r} \frac{dr}{d\phi} \right)^2 d\phi = n_r h \quad \text{--- ⑨}$$

~~or~~ ~~p_ϕ~~ ~~$\oint$~~ ~~$\left(\frac{1}{r} \frac{dr}{d\phi} \right)^2 d\phi$~~

The eqn of an ellipse in polar Co-ordinates

$$\frac{1}{r} = \frac{1 + \epsilon \cos \phi}{a(1 - \epsilon^2)} \quad \text{--- ⑩}$$

where ϵ is the eccentricity of the ellipse
 a & b = Semi major and Semi minor
 axes of the ellipse.



Taking log of both sides :-

$$-\log r = \log(1 + e \cos \phi) - \log a(1 - e^2)$$

Differentiating both sides w.r to ϕ :-

$$-\frac{1}{r} \cdot \frac{dr}{d\phi} = \frac{(0 - e \sin \phi)}{1 + e \cos \phi} - 0$$

$$\text{or } \frac{1}{r} \frac{dr}{d\phi} = \frac{e \sin \phi}{1 + e \cos \phi} \quad \text{--- (11)}$$

Putting eqn (11) in (9):

$$p_{\phi} \int \frac{e^2 \sin^2 \phi}{(1 + e \cos \phi)^2} d\phi = n_x h$$

$$\text{or } p_{\phi} e \int \frac{e \sin \phi}{(1 + e \cos \phi)^2} \cdot \sin \phi d\phi = n_x h$$

Integrating by parts:

$$p_{\phi} e \left[\sin \phi \int_0^{2\pi} \frac{e \sin \phi}{(1 + e \cos \phi)^2} d\phi - \int_0^{2\pi} \cos \phi \cdot \left(\frac{e \sin \phi}{1 + e \cos \phi} \right) d\phi \right] = n_x h$$



To find

$$\int_0^{2\pi} \frac{e \sin \phi}{(1 + e \cos \phi)^2} d\phi \quad \text{Putting } 1 + e \cos \phi = z$$

$$- e \sin \phi d\phi = dz$$

$$I = \int \frac{-dz}{z^2} = \frac{1}{z} = \frac{1}{1 + e \cos \phi}$$

$$p_\phi \in \left[\left\{ \int_0^{2\pi} \frac{e \sin \phi}{(1 + e \cos \phi)^2} d\phi \right\} \sin \phi - \int_0^{2\pi} \cos \phi \left\{ \frac{e \sin \phi}{1 + e \cos \phi} \right\} d\phi \right]$$

$$\therefore p_\phi \in \left[\left\{ \frac{\sin \phi}{1 + e \cos \phi} \right\}_0^{2\pi} - \int_0^{2\pi} \frac{\cos \phi d\phi}{1 + e \cos \phi} \right] = n_\phi h$$

$$- p_\phi \int_0^{2\pi} \frac{e \cos \phi d\phi}{(1 + e \cos \phi)} = n_\phi h$$

$$n - p_\phi \int_0^{2\pi} \left(1 - \frac{1}{1 + e \cos \phi} \right) d\phi = n_\phi h$$

$$n - p_\phi \left[\int_0^{2\pi} d\phi - \int_0^{2\pi} \frac{d\phi}{1 + e \cos \phi} \right] = n_\phi h$$

$$n - p_\phi \left[\right]$$

$$\star p_\phi \left[\frac{2\pi}{\sqrt{1-e^2}} - 2\pi \right] = n_\phi h$$



Putting eqⁿ ⑥

$$n_{\phi} h \left[\frac{1}{\sqrt{1-\epsilon^2}} - 1 \right] = n_r h$$

$$\therefore \frac{1}{\sqrt{1-\epsilon^2}} = \frac{n_r}{n_{\phi}} + 1 = \frac{n_r + n_{\phi}}{n_{\phi}} = \frac{n}{n_{\phi}} \quad \text{--- (12)}$$

where $n = n_r + n_{\phi}$ = an integer is known as the total Quantum number of the orbit.

$$b^2 = a^2 (1 - \epsilon^2)$$

$$\text{or } \frac{1}{\sqrt{1-\epsilon^2}} = \frac{a}{b} \quad \text{--- (13)}$$

Equating ⑫ and ⑬

$$\frac{a}{b} = \frac{n}{n_{\phi}} \quad \text{--- (14)}$$

We now find the total energy from ① :-

$$E_n = \frac{1}{2} m \left[\left(\frac{p_r}{m} \right)^2 + r^2 \left(\frac{p_{\phi}}{m r^2} \right)^2 \right] - \frac{z e^2}{4 \pi \epsilon_0 r}$$

$$E_n = \frac{1}{2} \frac{m}{m^2} \left[\left(\frac{p_{\phi} dr}{r^2 d\phi} \right)^2 + \frac{p_{\phi}^2}{r^2} \right] - \frac{z e^2}{4 \pi \epsilon_0 r}$$

$$E_n = \frac{1}{2} \frac{p_{\phi}^2}{m r^2} \left[\left(\frac{1}{r} \frac{dr}{d\phi} \right)^2 + 1 \right] - \frac{z e^2}{4 \pi \epsilon_0 r}$$



Putting Equations (10) and (11) :-

$$E_n = \frac{p_\phi^2}{2m a^2 (1-e^2)^2} \left[\frac{e^2 \sin^2 \phi}{(1+e \cos \phi)^2} + 1 \right] - \frac{Z e^2 (1+e \cos \phi)}{4\pi \epsilon_0 a (1-e^2)}$$

$$E_n = \frac{p_\phi^2 (1+e \cos \phi)^2}{2m a^2 (1-e^2)^2} \left[\frac{1+e^2+2e \cos \phi}{(1+e \cos \phi)^2} - \frac{Z e^2 (1+e \cos \phi)}{4\pi \epsilon_0 a (1-e^2)} \right] \quad (15)$$

Since the total Energy of the electron in the orbit remains same at every point, E_n should be independent of ' ϕ ' i.e. coefficient of $\cos \phi$ terms in eqⁿ (15) should be zero.

$$\frac{p_\phi^2 \cdot 2e}{2m a^2 (1-e^2)} = \frac{Z e^2 e}{4\pi \epsilon_0 a (1-e^2)} \quad (16)$$

$$\frac{p_\phi^2 \cancel{Z e}}{\cancel{2m} a^2 (1-e^2) \cancel{2}} = \frac{Z e^2 \cancel{e}}{4\pi \epsilon_0 \cancel{a} (1-e^2) \cancel{2}}$$

$$a = \frac{4\pi \epsilon_0 p_\phi^2}{Z e^2 m (1-e^2)}$$

Putting $p_\phi = \frac{n_\phi h}{2\pi}$

$$a = \frac{4\pi \epsilon_0 n_\phi^2 h^2}{Z e^2 m (1-e^2) \cdot 4\pi^2}$$

$$a = \frac{\epsilon_0 n_\phi^2 h^2}{\pi Z m e^2 (1-e^2)} = \frac{\epsilon_0 n^2 h^2}{\pi Z m e^2}$$



Sommerfeld Theory

$$\frac{a^2}{b^2} = \frac{1}{1-\epsilon^2}$$

$$\text{But } \frac{a^2}{b^2} = \frac{n^2}{n_{\phi}^2} = \frac{1}{1-\epsilon^2}$$

$$\therefore \frac{n_{\phi}^2}{1-\epsilon^2} = n^2$$

$$a = \frac{\epsilon_0 n^2 h^2}{\pi z m e^2} \quad (17)$$

$$\therefore b = \frac{\epsilon_0 n n_{\phi} h^2}{\pi z m e^2} \quad (18) \quad \left| \because b = a \frac{n_{\phi}}{n} \right.$$

Equations (17) and (18) gives the Semi-major and Semi minor axes of the ellipse.

The total Energy of the electron in the n th orbit can be written from eqⁿ (15) by omitting the coefficients of $\cos\phi$.

$$E_n = \frac{p_{\phi}^2 (1+\epsilon^2)}{2m a^2 (1-\epsilon^2)} - \frac{ze^2}{4\pi\epsilon_0 a (1-\epsilon^2)}$$

Using equation (16) in the above expression

$$E_n = \frac{ze^2 (1+\epsilon^2)}{\pi\epsilon_0 a (1-\epsilon^2)} - \frac{ze^2}{4\pi\epsilon_0 a (1-\epsilon^2)}$$

$$E_n = - \frac{ze^2}{8\pi\epsilon_0 a (1-\epsilon^2)} [-1 - \epsilon^2 + 2]$$

$$\text{or } E_n = - \frac{ze^2 (1-\epsilon^2)}{8\pi\epsilon_0 a (1-\epsilon^2)} = - \frac{ze^2}{8\pi\epsilon_0 a}$$

Putting equation (7) :-

$$E_n = - \frac{ze^2}{8\pi\epsilon_0 \cdot \frac{\epsilon_0 n^2 h^2}{\pi z m e^2}} = - \frac{1}{8\epsilon_0^2} \frac{z^2 m e^4}{n^2 h^2} \quad (19)$$



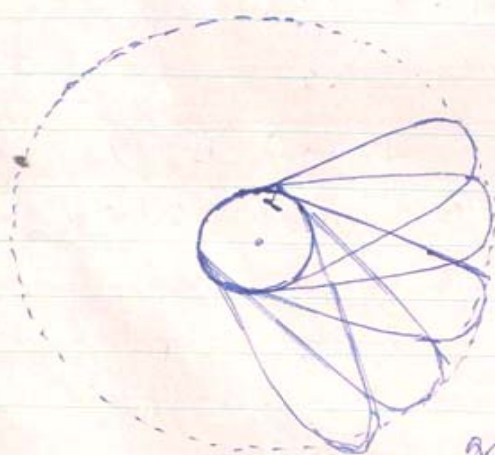
Sommerfeld also introduced the relativistic variation of mass due to variation of speed of the electron, at different points in the orbit. The final expression for energy in terms of wave no. was shown to be

$$\frac{E_n}{hc} = - \frac{RZ^2}{n^2} - \frac{R\alpha^2 Z^4}{n^2} \left(\frac{1}{n_\phi} - \frac{3}{4n} \right) \quad (20)$$

where α = fine structure constant = $\frac{e^2}{Z \hbar c}$ $\left| m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}} \right|$

For the same quantum numbers n , n_ϕ can have 'n' values, for each of them energy will be slightly different.

It is found that the path of the electron is no longer a simple ellipse but is a precessing ellipse and the complicated curve generated by a precessing ellipse is known as Rosette.



$$c = \lambda \nu$$

$$E_n = h\nu = h \frac{c}{\lambda} = hc\bar{\nu}$$

$$\bar{\nu} = \frac{E}{hc}$$